

INPhO 2005

1. Javelin Throw

[8]

- (a) A projectile is thrown in the forward direction with speed u (relative to the vehicle) and angle θ from a vehicle moving on a horizontal surface with speed v . The motion of the vehicle and the projectile is in the same plane. Obtain an expression for the range R .
- (b) Obtain an expression for the angle $\theta_{\max}$ for which R is maximum. Examine if $\theta_{\max}$ is less than or greater than 45° .
- (c) An athlete moving at 5 m/s throws a javelin at 20 m/s to achieve maximum range. What is the angle of the athlete's throw? Briefly comment on your result.

2. Rolling Sphere

[8]

A sphere of radius R is projected up an inclined plane of angle θ with initial speed v_0 and angular velocity ω_0 in a direction in which it will roll up. The coefficient of friction is $\tan \theta/7$ and it is given that $v_0 > R\omega_0$. Give the direction of friction for different parts of motion as the sphere rises. Obtain an expression for the total time of motion of the sphere up the plane before it stops.

3. Vibrating Rod:

[6]

A metallic rod of length 1 m is rigidly clamped at its mid-point. Longitudinal stationary waves are set up in the rod such that there are two nodes each on either side of the mid-point. The amplitude of an antinode is 10^{-6} m. Write the equation of motion at a point 0.1 m from the mid-point and the equations for the component travelling waves in the rod. Take the origin to be at mid-point of the rod. The density of the rod is 8000 kg/m^3 and its Young's modulus is $2 \times 10^{11} \text{ N/m}^2$.

4. Heat and Work in a Double Chamber

[8]

Consider a double chambered vessel with thermally insulated walls. There is a frictionless, thermally insulating movable partition between the two chambers. On each side, there are n moles of an ideal gas that is initially at pressure P_0 , volume V_0 , and temperature T_0 . The heater in the first chamber supplies heat very slowly. The gas expands until the pressure in the second chamber is $27/8 P_0$. The ratio of the specific heat at constant pressure (c_p) to the specific heat at constant volume (c_v) γ for the gas is 1.5, and c_v is independent of temperature. In terms of n , c_v , and T_0 ,

- (a) How much work is done on the gas in the right chamber?
- (b) What is the final temperature of gas on the right?
- (c) What is the final temperature of the gas on the left?
- (d) How much heat flows into the gas on the left?

5. Analyzing an Image

[6]

The distance between a screen and a light source lined up on optical bench is 120 cm. When a lens is moved along the line joining them, sharp images of the source can be obtained at two lens positions; the size ratio of these two images is 1:9.

- (a) What is the focal length of lens?
- (b) Which image is brighter?
- (c) Determine the ratio of the brightness values of the two images.

6. The Star ζ Puppis

[6]

- (a) In 1896 Pickering observed mysterious lines from the star ζ Puppis. He fit their wavelengths λ to the empirical formula:

$$\frac{1}{\lambda} = R \left(\frac{1}{\left(\frac{n_1}{2}\right)^2} - \frac{1}{\left(\frac{n_2}{2}\right)^2} \right)$$

where R is the Rydberg constant and n_1 and n_2 are integers. Identify the species responsible for this spectrum.

- (b) Identify any three lines in the visible part of the spectrum [3800 – 7000 Å].

7. The Photoelectric Effect for Zinc:

[10]

Light of net intensity $1.0 \mu\text{W}/\text{cm}^2$, falls on a clean sample of zinc of area 1.0 cm^2 . Zinc reflects 96% of light and only 3.0% of the absorbed light lies in the violet range of the spectrum above the threshold frequency.

What is the intensity of light actually available for the photoelectric effect?

- (a) Assuming that all the photons in the violet range have an effective wavelength of 250 nm ($1 \text{ nm} = 10^{-9} \text{ m}$), how many electrons are emitted per second? Assume 100% efficiency of photoelectric emission.
- (b) Estimate the current in the phototube.
- (c) If the cut-off wavelength is 288 nm, find the work function of zinc.
- (d) Find the stopping voltage if photoelectrons are produced by light with wavelength 250 nm.

8. Rotation of the Earth

[10]

The rotation of the earth is responsible for many unusual and fascinating effects. Here we shall consider three of them.

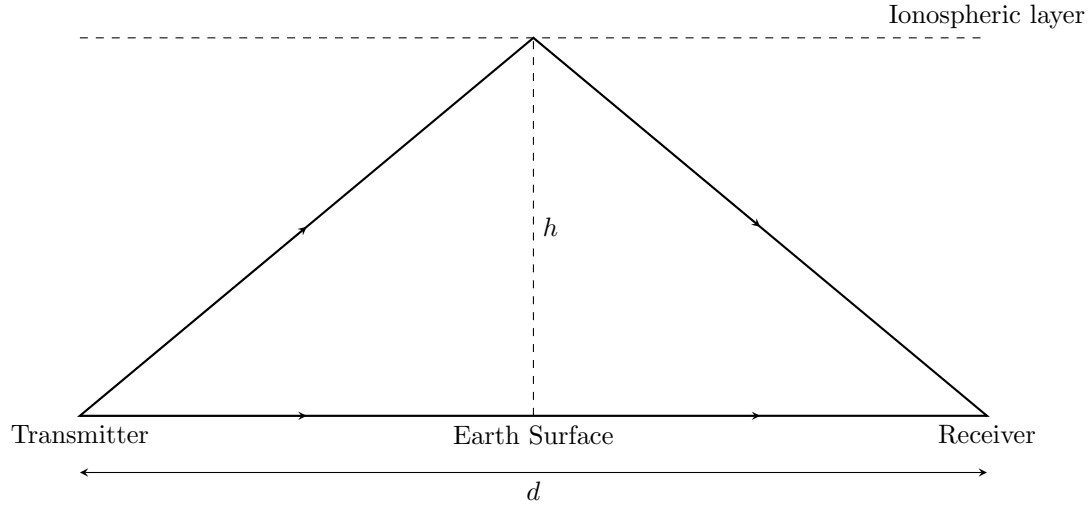
- (a) *Eastward deflection of falling object.*
Consider a body dropped from height $h = 1000 \text{ m}$ on the equator. Because of rotation of earth, the body will deflect towards east. Estimate this deflection. (Ignore air resistance)
- (b) Consider a pendulum suspended “vertically” at latitude λ . Obtain an expression for the apparent weight of the bob of the pendulum and the angle made by the pendulum with the vertical. At what latitude will this angle be the maximum?
- (c) Although the rotation of the earth has a period of 24 hours, alternate high tides occur after approximately 25 hours. Provide a simple explanation for this anomaly.

9. Ionospheric Reflection

[6]

A short wave radio receiver receives simultaneously two signals of frequency 10 MHz from a transmitter 600 km away, one by a path along the surface of earth, and one by reflection from the portion of the ionospheric layer situated at a height of 250 km. The layer acts as a perfect horizontal reflector. The ionosphere layer is moving with a slow vertical speed of 3 m/s. Ignore Doppler shift and the curvature of the earth.

- (a) Explain qualitatively why the receiver gets a signal with fluctuating intensity.
- (b) Obtain a quantitative estimate of this fluctuation.



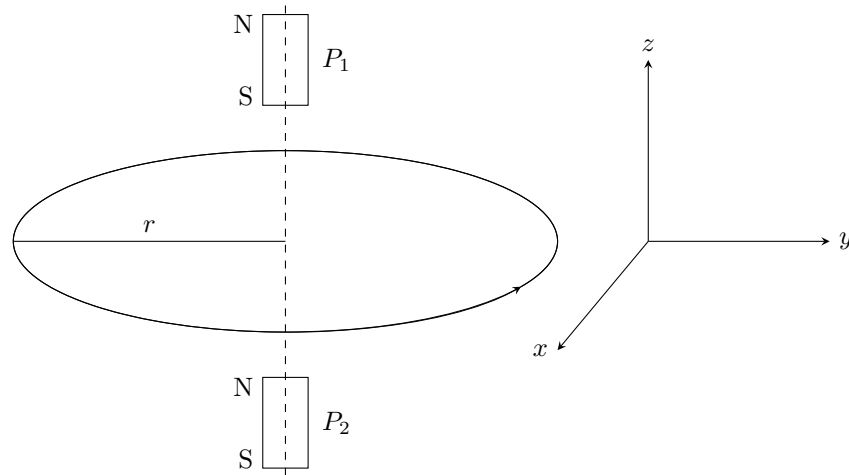
h - Height of the reflecting surface

d - Length of the direct path along earth's surface

10. Suspended Magnet

[10]

A ring of radius r , carrying current i , is placed in the horizontal xy plane ($z = 0$) with its axis along z -direction as shown in figure. A short bar magnet of moment μ and mass m is placed on the axis such that it is in stable equilibrium.



- Give the expression for magnetic field $B(z)$ at an arbitrary axial point z due to the current loop.
- Sketch the first and second derivatives of magnitude of magnetic fields $B'(z)$ and $B''(z)$ respectively. (Only sketch is required, no need to draw the graph precisely.)
- Which of the positions P_1 and P_2 of the bar magnet can possibly be an equilibrium position?
- Based on the sketches of part (b) argue what one would expect if the bar magnet is displaced vertically from the equilibrium position by a small amount?
- It is found that the bar magnet is in equilibrium at position $|z| = r$. Given $r = 0.1$ m, and $i = 10$ A, and $\mu = 2.95 \times 10^4$ A-m² obtain the mass m .

11. Birth of a Star

[12]

We wish to estimate the minimum mass that a gaseous hydrogen cloud contracting under its gravity should have to become a star. Consider the cloud to have a spherical shape with volume Ω and mass M distributed uniformly.

- (a) Show that the gravitational pressure of the cloud is

$$P_G(\Omega) = -\frac{1}{5} \left(\frac{4\pi}{3} \right)^{\frac{1}{3}} \frac{GM^2}{\Omega^{\frac{4}{3}}}$$

- (b) In equilibrium, the inward gravitational pressure is counterbalanced by the outward pressure in the clouds. This outward pressure is the sum of the usual gaseous pressure and the so called electron “degeneracy pressure”. The latter is a quantum mechanical effect arising from the Pauli’s exclusion principle from electrons. The total electronic energy is

$$E_e = \frac{\pi^3 \hbar^2 N_e^{5/3}}{10 m_e \Omega^{2/3}} \left(\frac{3}{\pi} \right)^{5/3}$$

where N_e is the total number of electrons in the cloud and the other symbols have their usual meaning. Determine the electron “degeneracy pressure” from the above expression. Let $\Omega = N_e d^3$ where d is the average distance between the protons. Obtain an expression for the maximum temperature in terms of m_e , m_p and M that a cloud can attain. Note m_e and m_p are respectively the masses of electron and proton. Assume that the cloud is an ideal gas.

- (c) A contracting cloud becomes a star when the temperature is sufficient to initiate a nuclear reaction. Assume that fusion is initiated when two protons “touch” each other. Take the radius of the proton to be $r = 10^{-15}$ m. Estimate the minimum temperature needed to trigger nuclear fusion reaction.
- (d) The minimum temperature thus obtained is two orders of magnitude larger than the actual temperature for fusion to set up. Give two reasons why we have overestimated the temperature in part (c).
- (e) From (b) and (c) estimate the minimum mass a gaseous cloud must possess to transform into a star.